

tion. There exists a significant opportunity for this industry, particularly in telecom equipment, consumer electronics, medical devices and solar photovoltaics. The soaring demand for electronics, together with a favourable policy environment, presents the electronics sector with a golden chance for revival.

However, if we consider the wide range of customer choices in any industry today, there is a requirement of fast and constantly changing technology. Owing to competitive pressure, this certainly leads to a reduction in product costs.

Component manufacturers are expected to work in a smart manner with high efficiency to match product life cycles with the changing customers' needs. The idea is to support them in the market, while constantly upgrading the technology and operational excellence.

Challenges in India

Like any other industry, the connectors and terminals industry also experiences many challenges due to increasing demands of the end users. India needs to urgently fix several disadvantages in the manufacturing ecosystem.

Despite having skilled engineers, India has an acute shortage of skilled labourers, and its inverted tax structure makes domestic manufacturing unviable. For sustainable competitive advantage, manufacturing companies across the sector have increased focus on R&D and innovation in the last three years.

More initiatives from Indian government are yet to come, like the Russian initiatives to promote high-tech innovations, such as Rusnano—the nanotechnology hub. The designers in the industry are currently in the nascent stage of customisation and enhancements, and minimal new product development. Thus, in order to evolve as a design hub, we have quite a long way to go.

The other challenge to this market in India comes from the manufacturers and distributors in China and Taiwan who dominate with cheap solutions and prices. Hence, new manufacturing units as well as the existing manufacturers are in trouble due to these suppliers.

"Even for FCI, technologically ad-

Forecasts

Asia-Pacific (APAC) is forecast to witness the highest growth of 5.4 per cent CAGR by 2017. Steady economic growth and expanding middle class population aspiring for a better lifestyle in China and India indicate substantial potential for growth in APAC, says the global automotive and transportation connector industry report by Lucintel.

Global Information, Inc. forecasts in its connector industry forecast published on November 30, 2013 that 2014 sales will grow by 6.7 per cent.

The demand for electronic products in India—for domestic consumption and exports—is projected to grow at 22 per cent per annum, to reach US\$ 400 billion by 2020, as per MAIT. At current growth levels of 16 per cent, it is estimated that the country's trade deficit because of electronic products could exceed that of oil products by 2020.

Technology trends

Connectors are increasingly becoming smaller, and those for consumer electronic devices are being designed for ease of use. For instance, USB Implementers Forum has announced the new USB type-C connector. It is similar to a microUSB connector but at the same time is reversible and asymmetrical as well.

As per Priya Kuber, managing director, Arduino India, "Industry of connectors has seen many steps to create standardisation. In developing Arduino and its clones, it has been noted that the manufacturers are looking for more easy availability and more and more people are tending to use microUSB."

Nano miniature connectors

These connectors reduced in size and weight with approximately 0.0635cm pitch. These are the special range of connectors for space programmes that are inspected as per EEE-INST-002. "Nano miniature connectors offer critical size and weight savings for rugged environments, deliver high reliability, low contact resistance and wide operating temperature range. These are ten times smaller in volume and about 10 per cent of the weight of a standard micro-D connector," says Sanjay Handu, director, Aerospace, Defense & Marine - TE Connectivity, India.

Customised IP-67-rated connectors

Though not new in the market, they are becoming increasingly more common because of their customised configurations and industry standards. Their main application areas are the ones that need high level of protection from the elements such as future soldier systems, GPS-related equipment and handheld military tactical radios.

High-speed connectors

With the increasing usage of next-generation telecom initiatives, such as 4G/LTE, the telecom segment is moving towards very high-speed connectors. "FCI has a very strong presence in the communications sector and we have introduced many high-speed connectors over this period, such as Milipacs, Metralex and AirMax VS, which support data transfer rates ranging from 4Gbps to 25Gbps. Some of these connectors are designed and manufactured at our design and engineering centre in Cochin," says Alex Perrotta, CEO, FCI Electronics and Rajesh Arakkal, general manager, FCI OEN India in an interview with *Electronics Bazaar* magazine.

vanced products are sold in a much larger volume abroad, compared to those sold in India. However, some of our core products, such as wire-to-board, board-to-board connectors and PCB interconnects, are sold in high volumes in India," says Rajesh Arakkal, general manager, FCI OEN Connectors.

According to Rajesh, "Although the government's initiative to promote domestic manufacturing of 25 electronic products is a welcome boost to the lacklustre Indian economy, we hope the government will continue to abide by its preferential market access

policies, and ensure its 'Made in India' slogan is well executed."

Price dependency on the raw materials

Cost remains one of the most sensitive factors for standard products having a high volume of sale. The pricing of the connectors varies for each application. Several factors, such as quality, specifications and volume of production, determine the cost of a product. But considering various factors, such as exchange rate and increasing raw material cost, it has been a very challenging